

Cable Metamaterial Design Project

Vibration Control Using Metamaterials

Exercise Description

This project focuses on designing a metamaterial cable with local resonators to suppress vibrations at the fourth mode shape. The cable is subjected to varying axial tension (at least two different axial loads should be considered), simulating operating conditions in which tension may fluctuate due to thermal effects, operational loads, or environmental factors. The cable has a length of $L = 51.95$ m, cross-sectional area $A = \pi(0.02515/2)^2$ m², Young's modulus $E = 203$ GPa, density $\rho = 3000$ kg/m³, and nominal tension $T_0 = 1.68 \times 10^4$ N [2, 1].

Students will design resonators tuned to the fourth-mode frequency, using a total resonator mass ratio of less than 20% of the cable mass, distributed among 10 or more resonators. They will then analyse the receptance and dispersion curve of the metamaterial cable, while investigating how variations in axial tension from 70% to 130% of the nominal value affect its performance. The design must achieve at least a 15 dB reduction in the fourth-mode response at the nominal tension, maintain a reduction of at least 12 dB across the specified tension range, and keep the total resonator mass below 20% of the cable mass. The expected results are a 15–20 dB reduction at the target frequency, with bandgaps forming around the controlled mode frequency.

Key Parameters

Parameter	Value
Cable Length	51.95 m
Cross-sectional Area	$\pi(0.02515/2)^2$ m ²
Young's Modulus	203 GPa
Density	3000 kg/m ³
Nominal Tension	1.68×10^4 N
Target Frequency	10 Hz
Mass Ratio	15%
Number of Resonators	Select the number
Minimum Reduction Required	15 dB

Table 1: Design parameters for the cable metamaterial exercise.

Design Requirements

- Minimum reduction at 10 Hz: 15 dB
- Performance across $\pm 20\%$ tension variation: >12 dB
- Total resonator mass: $<20\%$ of cable mass

The cable is modelled using Euler-Bernoulli beam theory with axial force as in [2, 1]. When a resonator is attached, the effective response is modified by the relative mass and can be analytically computed as in [3, 1].

References

- [1] Machado, M.R. & Dutkiewicz, M. (2024). "Enhancing broadband vibration suppression of a cable conductor using graded metamaterials." *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46(4).
- [2] M.R. Machado, M. Dutkiewicz, C.F.T. Matt, D.A. Castello, Spectral model and experimental validation of hysteretic and aerodynamic damping in dynamic analysis of overhead transmission conductor, *Mechanical Systems and Signal Processing*, 136, 2020, 106483.
- [3] C. Sugino, Y. Xia, S. Leadenham, M. Ruzzene, A. Erturk, A general theory for bandgap estimation in locally resonant metastructures, *Journal of Sound and Vibration*, 406, 2017, 104-123, <https://doi.org/10.1016/j.jsv.2017.06.004>.